

ASTRODYNAMICS I: HOMEWORK #3

Due: Wed, May 2, 11:00AM, 2018

Read carefully pp. 151-165 of Vallado (Section 3.3 Coordinate Systems).

1. Briefly summarize the following terminology:
 - A. Great Circles
 - B. Celestial Poles, Celestial Equator
 - C. Hour Circles
 - D. Ecliptic
 - E. Line of Nodes, Vernal Equinox, Autumnal Equinox
 - F. Ecliptic (Celestial) Latitude & Longitude
 - G. Right Ascension and Declination
 - H. Hour Angle, Local Hour Angle, Greenwich Hour Angle
2. Briefly summarize all Coordinate Systems in Section 3.3. You don't have to discuss in detail. Describe succinctly the origin, the direction of each axis, and key characteristics of them.

Read carefully pp. 179-205 of Vallado (Section 3.5 Time).

3. Discuss the following briefly:
 - A. Main purpose of 'time' (according to Newcomb)
 - B. Epoch
 - C. Determination of an epoch requires that:
 - D. What kinds of time interval exist?
4. Specify four main time scales and discuss their characteristics briefly.

5. Regarding solar time, discuss the following terminologies briefly.
 - A. Apparent Solar Time
 - B. Mean Solar Time, Fictitious Mean Sun (FMS)
 - C. Universal Time (UT), UT0, UT1, UT2
 - D. Equation of Time (EQ_{Time})
 - E. Coordinated Universal Time (UTC)
 - F. *Julian Date (JD) – *please describe in detail; JD is a key concept in Astrodynamics.*
6. Regarding sidereal time, discuss the following terminologies briefly.
 - A. Greenwich Mean Sidereal Time (GMST)
 - B. Local Sidereal Time (LST)
 - C. Mean Sidereal Time (MST)
 - D. Apparent Sidereal Time (AST)
 - E. Equation of Equinoxes
7. Describe the procedure of computing GMST, LST from JD.
8. Regarding dynamical time, discuss the following terminologies briefly:
 - A. Terrestrial Time (TT), which used to be Terrestrial Dynamical Time (TDT)
 - B. Barycentric Dynamical Time (TDB)
 - C. Geocentric Coordinate Time (TCG)
 - D. Barycentric Coordinate Time (TCB)

Note that I do not ask you to make a fancy report! I do ask you to study Section 3.3 and 3.5 by yourself because it is the best way to understand this section. You may simply submit an informal report as evidence.